
Main Title
Class subtitle

Class subtitle

07/09/2025 - 01/07/2026

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Main title

I - Maths

For my maths class, I made these things:

I.I - #definition

Definition 1.1. (Linearity):

We say that φ is linear (homomorphism) if:

$$\varphi(\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_n X_n) = \lambda_1 \varphi(X_1) + \lambda_2 \varphi(X_2) + \dots + \lambda_n \varphi(X_n) \quad (1.1.1.1)$$

I.II - #example

Example 1.1. (Example title): Basic text.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat.

$$\begin{aligned} \varphi(0, 0, 0) &= (0, 0) = 0_{\mathbb{R}^2} \\ \varphi(\alpha X_1 + \beta X_2) &\stackrel{?}{=} \alpha \varphi(X_1) + \beta \varphi(X_2) \end{aligned} \quad (1.1.2.2)$$

I.III - #theorem

I.III.1 - With title

Theorem 1.1. (Stokes' Theorem):

Let M be an oriented differential manifold with boundary of dimension n , and ω a $(n-1)$ -form differential form with compact support on M of class C_1 .

Then, we have:

$$\int_M d\omega = \int_{\{\partial M\}} i^* \omega \quad (1.1.3.3)$$

where d denotes the exterior derivative, ∂M the boundary of M , equipped with the induced orientation,

and $i^* \omega = \omega|_{\{\partial M\}}$ the restriction of ω to ∂M .

I.III.2 - Without title

Theorem 1.2.

Let E be a finite-dimensional vector space, F a vector subspace of E , and $B = (X_1, X_2, \dots, X_n)$ a basis of F .

Then, there exists a basis $(X_1, X_2, \dots, X_n, X_{\{n+1\}}, \dots, X_m)$ of E such that $(X_1, X_2, \dots, X_n)$ is a basis of F .

I.IV - Custom styling

Definition 1.2. (**Styled Definition**): A **group** is a set G equipped with a binary operation $\cdot$ satisfying closure, associativity, identity, and invertibility.

Theorem 1.3. (**Styled Theorem**):

For any right triangle with sides a , b , and hypotenuse c :

$$a^2 + b^2 = c^2 \quad (1.1.4.4)$$

I.V - ar

For vectors, I use `ar(X)` and it gives $\vec{X}$.

II - Subtitle

II.I - Subsubtitle

Custom Block

Styled Block

This block uses custom title and body styling.

Custom Blockquote

Basic inline raw text

This code block uses `#code()` macro.

src/string_utils.rs | rust

```
1 /// Extension traits and utilities for string manipulation
2 ///
3 /// This module provides additional functionality for working with strings,
4 /// including title case conversion and other string transformations.
5 use std::string::String;
6
7 /// Trait that adds title case functionality to String and &str types
8 pub trait TitleCase {
9     /// Converts the string to title case where each word starts with an uppercase letter
10     /// and the rest are lowercase
11     ///
12     fn to_title_case(&self) -> String;
13 }
14
15 impl TitleCase for str {
16     fn to_title_case(&self) -> String {
17         self.split(|c: char| c.is_whitespace() || c == '_' || c == '-')
18             .filter(|s| !s.is_empty())
19             .map(|word| {
20                 // If the word is all uppercase and longer than 1 character, preserve it
```

```
21         if word.chars().all(|c| c.is_uppercase()) && word.len() > 1 {
22             word.to_string()
23         } else {
24             let mut chars = word.chars();
25             match chars.next() {
26                 None => String::new(),
27                 Some(first) => {
28                     let first_upper = first.to_uppercase().collect::<String>();
29                     let rest_lower = chars.as_str().to_lowercase();
30                     format!("{}", first_upper, rest_lower)
31                 }
32             }
33         }
34     })
35     .collect::<Vec<String>>()
36     .join(" ")
37 }
38 }
39
40 impl TitleCase for String {
41     fn to_title_case(&self) -> String {
42         self.as_str().to_title_case()
43     }
44 }
45
46 #[cfg(test)]
47 mod tests {
48     use super::*;
49
50     #[test]
51     fn test_title_case_str() {
52         assert_eq!("hello world".to_title_case(), "Hello World");
53         assert_eq!("HASH_TABLE".to_title_case(), "HASH TABLE");
54         assert_eq!("dynamic-programming".to_title_case(), "Dynamic Programming");
55         assert_eq!("BFS".to_title_case(), "BFS");
56         assert_eq!("two-sum".to_title_case(), "Two Sum");
57         assert_eq!("binary_search_tree".to_title_case(), "Binary Search Tree");
58         assert_eq!(" spaced words ".to_title_case(), "Spaced Words");
59         assert_eq!("".to_title_case(), "");
60     }
61 }
```